

accuracy elec. structure

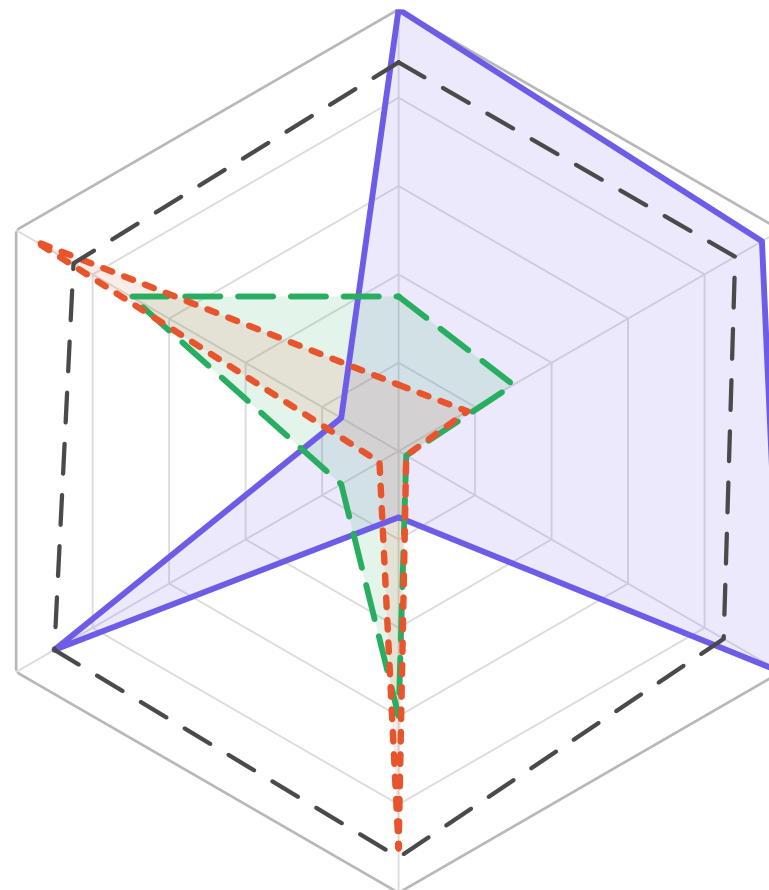
force field $\rightarrow$ multireference

degrees of freedom

$2 - 3 \rightarrow 10^5$

accuracy nuclei dynamics

classical $\rightarrow$ semi-classical $\rightarrow$ quantum



light-matter interaction

implicit $\rightarrow$ explicit field

electron dynamics

no $\rightarrow$ yes

time scale of the process

fs $\rightarrow$ ns

— quantum / ab initio

— classical MD

- - - coarse-grained

- - - what the problem demands